Date:

Subject: قبل الميد ترم بأسبوع   Quiz (1)   موضوع الدرس:

Question (1):

a) Show that The DFT, $X(k)$, of a sequence $x(n)$ is periodic.

b) in A/D, in detail, explain The quantization process of Continous amplitude signals.

c) For The signal $x(t) = 2\cos(200\pi t)$ to be sampled with sampling freq $F_s = 400\text{Hz}$.

(i) Find an expression For its Corresponding sequence $x(n)$, Then Calculate fundemental period N_p.

(ii) Find The samples value of The first Cycle in $x(n)$ Then plot Them at The same figure with $x(t)$.

Question (2):

a) Deduce The basic relation between The Fourier transform and The discrete Fourier transform.

b) in detail. Find The Characterists of The Continous time Sinusoidal signal?

c) For The following sequence: $x(n) = \{1, 0, 1, 1\}$

i) using The FFT method find The frequency Component of $x(n)$, $X(k)$.

ii) check your results in (i) by Calculating $x(n)$ using IDFT

iii) using your results, verify parseval's Theorem.

iv) plot The amplitude and The phase spectrum.

v) what is The Computational saving when using FFT instead of DFT?

Date: / /

Subject: The Answer of Quiz

Question (1)

a) discrete Fourier transform Convert A periodic signal at time domain to Periodic signal at Freq domain

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-jnk\frac{2\pi}{N}}$$

to prove that put $K = K + Np$

$$X(K+N) = \sum_{n=0}^{N-1} x(n)\, e^{-jn\frac{2\pi}{N}(K+N)}$$

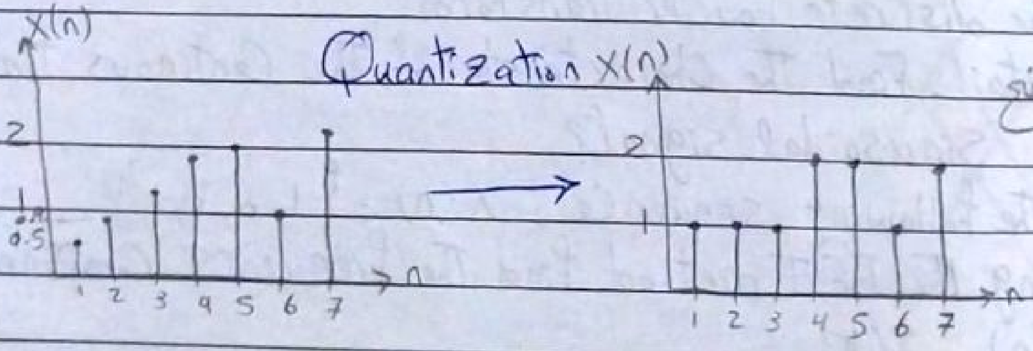
$$X(K+N) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}nK - j2\pi n}$$

$$e^{x+y} = e^{x} \cdot e^{y}$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}nk} \cdot \cancel{e^{-j2\pi n}} = 1$$

$$\therefore X(K+N) = X(K) \quad \text{periodic.}$$

b)



الرسم موضح ✓

Convert signal after sampling from Continous Amplitude to discrete Amplitude by Approximate values to The nearest integer number.

c) $X(t) = 2\cos(200\pi t)$

$x(t) = A\cos(2\pi f t)$ هقارنها بالصوره العامة

A (Amplitude) $= 2$

$2\cancel{\pi} f \cancel{t} = 200\cancel{\pi}\cancel{t}$ $2f = 200$ $\boxed{f = 100}$

$F_s = 400$ $F = \frac{f}{F_s} = \frac{100}{400} = \frac{1}{4}$ $N_p = 4$

expression for $x(n)$: $\boxed{x(n) = 2\cos(200\pi n t_s)}$

$t_s = \frac{1}{F_s} = \frac{1}{400} = \underline{2.5 * 10^{-3}}$

Fundemental period, $\boxed{N_p = 4}$

→ we have 4 samples so we will get it

$X(n) = 2\cos(200\pi * 2.5 * 10^{-3} n)$

at $n = 0$

$X(0) = 2\cos(0) = 2$

at $n = 1$ →

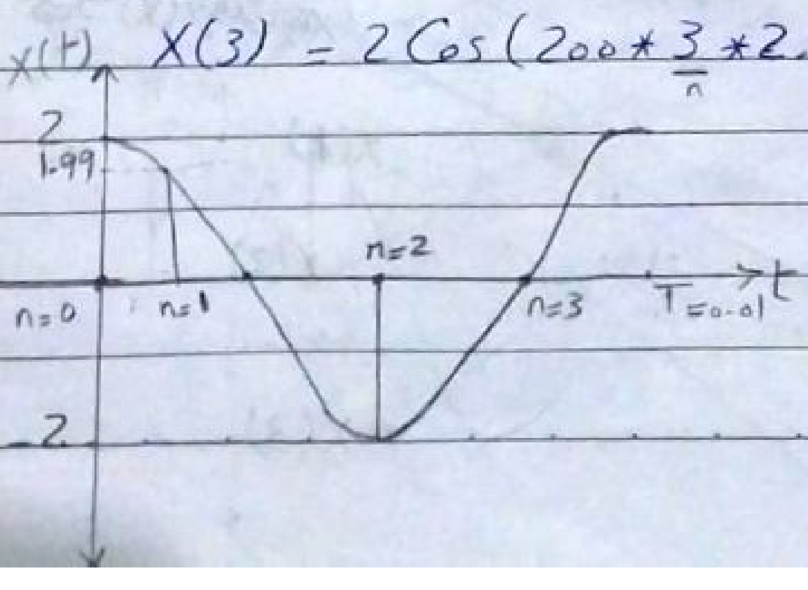
$X(1) = 2\cos(200 * 2.5 * 10^{-3}\pi) = 1.99$

at $n = 2$

$X(2) = 2\cos(200 * \underset{n}{2} * 2.5 * 10^{-3}\pi) = -2$

at $n = 3$

$X(3) = 2\cos(200 * \underset{n}{3} * 2.5 * 10^{-3}\pi) = 0$



هنجيب الـ T من f

$T = \frac{1}{f} = \frac{1}{100} = 0.01$

زمن one cycle

أضرب كدا إنه الـ (Cos) Signal

يعني يتبدأ من القمة هنا

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Question (2)

a) Fourier transform used to Convert A periodic Continous Time signal to periodic Continous signal at frequency domain

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

* used integral *

Discrete Fourier transform used to Convert A periodic discrete in time domain to periodic discrete in frequency domain

$$\rightarrow X(k) = \sum x(n)\, e^{-jn \frac{2\pi}{N} k}$$

* using Summission *

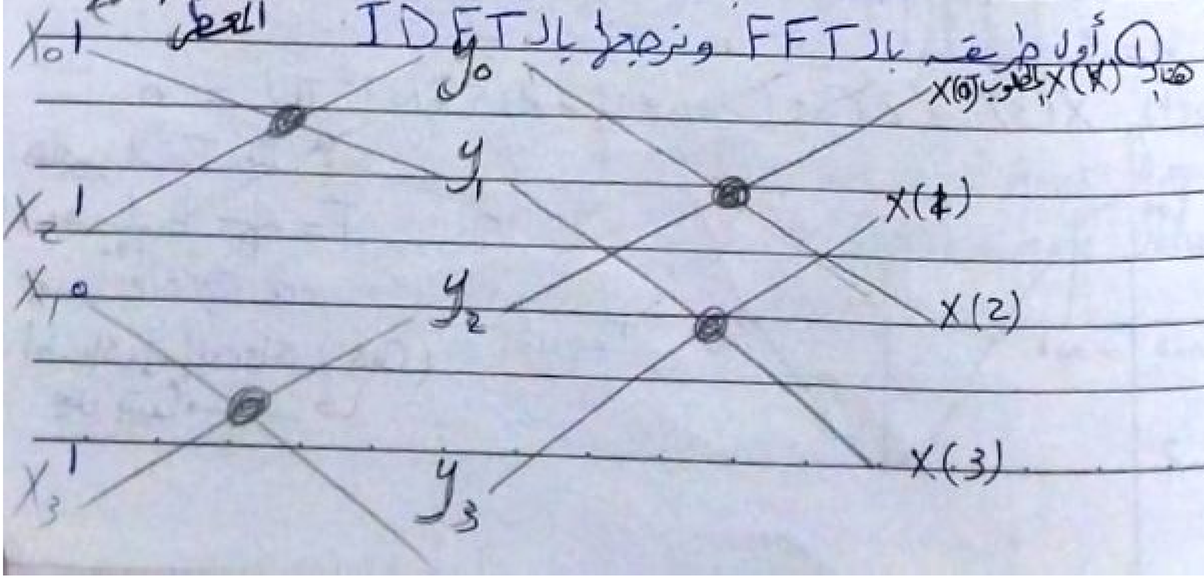
b) The Characteristic in Sinusoidal signal is all of analog signal are a group of sin signal any analog signal = Σ Sinwave signal. So we Can deal or process on any analog signal (Sinusoidal is easy to doing process on it)

c) هنحل بـ أنه دي مرتين ← هنجيبها بال FFT ونرجعها بـ IDFT
هنجيبها بال DFT ونرجعها بـ IFFT

علشانه بس الامتحان

X(n) المعطى

① أول طريقة بال FFT ونرجعها بال IDFT



$y_0 = x_0 + x_2 = 1 + 1 = 2$

$y_1 = x_0 - x_2 = 1 - 1 = 0$

$y_2 = x_1 + x_3 = 0 + 1 = 1$

$y_3 = x_1 - x_3 = 0 - 1 = -1$

spectrum $X(K)$:

$X(0) = y_0 + y_2 = 2 + 1 = 3$

$X(1) = y_1 - Jy_3 = 0 - J*-1 = J$

$X(2) = y_0 - y_2 = 2 - 1 = 1$

$X(3) = y_1 - (-J)y_3 = 0 - (-J*-1) = -J$

$\therefore$ spectrum $X(K) = \{\overset{X(0)}{3}, \overset{X(1)}{J}, \overset{X(2)}{1}, \overset{X(3)}{-J}\}$

هنرجع بقا بعد عاوزين نجيب $x(n)$ اللي هو معطى
بس بطريقة بال IDFT يعني هنتأكد من الحل

$$\longrightarrow x(n) = \frac{1}{N}\sum_{n=0}^{N-1} X(K)\, e^{JnK\frac{2\pi}{N}} \qquad \boxed{N=4}$$

دايما كده عشان K من 0

الصورة العامة

$$\Rightarrow x(n) = \frac{1}{4}\sum_{K=0}^{3}\Big[X(0)e^{0} + X(1)e^{Jn\overset{\frac{\pi}{2}}{\frac{2\pi}{4}}} + X(2)e^{Jn\frac{2\pi}{4}*2 \to \pi} + X(3)e^{Jn\frac{2\pi}{4}*3 \to \frac{3\pi}{2}}\Big]$$

at $n=0$

$$x(0) = \frac{1}{4}\Big[3*1 + J*\overset{1}{e^{J*0*\frac{\pi}{2}}} + 1*\overset{1}{e^{J*0*\pi}} + (-J)*\overset{1}{e^{J*0*\frac{3\pi}{2}}}\Big] = 1$$

at $n=1$

$$x(1) = \frac{1}{4}\Big[3*1 + \overset{-1}{J*\overset{J}{e^{J\frac{\pi}{2}}}} + \overset{-1}{1*\overset{-1}{e^{J\pi}}} + \overset{-1}{(-J)*\overset{-J}{e^{J\frac{3\pi}{2}}}}\Big] = \frac{1}{4}*0 = 0$$

$3 - 1 - 1 - 1$

at $n=2$

$$x(2) = \frac{1}{4}\Big[3*1 + J*\overset{-1}{e^{J\pi}} + 1*\overset{1}{e^{J2\pi}} + (-J)*\overset{-1}{e^{J3\pi}}\Big] = \frac{1}{4}*4 = 1$$

$3 \quad -J \quad +1 \quad +J$

$$x(3) = \frac{1}{4}\Big[3*1 + J*\overset{-J}{e^{J\frac{3\pi}{2}}} + 1*\overset{-1}{e^{J3\pi}} + (-J)\overset{J}{e^{J\frac{9\pi}{2}}}\Big] = \frac{1}{4}*4$$

$3 \quad +1 \quad -1 \quad +1$

$x(3) = 1$

كده حلها صح $x(n) = \{\overset{\circ}{1}, 0, 1, 1\}$:

② الطريقه التانيه هجيب بال DFT وترجع بال IFFT

DFT بتقول

$$\Rightarrow X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-Jnk\frac{2\pi}{N=4}}$$

الصوره العامه

$$\rightarrow X(k) = \Big[\overset{1}{\cancel{x(0)}}\, e^{0} + \overset{0}{\cancel{x(1)}}\, e^{-Jk\frac{\pi}{2}} + \overset{1}{\cancel{x(2)}}\, e^{-Jk\pi} + \overset{1}{\cancel{x(3)}}\, e^{-Jk\frac{3\pi}{2}}\Big]$$

$$X(k) = \Big[1 + 0 + e^{-Jk\pi} + e^{-Jk\frac{3\pi}{2}}\Big]$$

سهله خالص أهو، أعوض بقا

at $k=0$

$$\rightarrow X(0) = \big[1 + 0 + \overset{1}{\cancel{e^{0}}} + \overset{1}{\cancel{e^{0}}}\big] = 3$$

at $k=1$

$$\rightarrow X(1) = \Big[\cancel{1} + 0 + \overset{\cancel{1}}{\cancel{e^{-J\pi}}} + \overset{J}{\cancel{e^{-J\frac{3\pi}{2}}}}\Big] = +J$$

at $k=2$

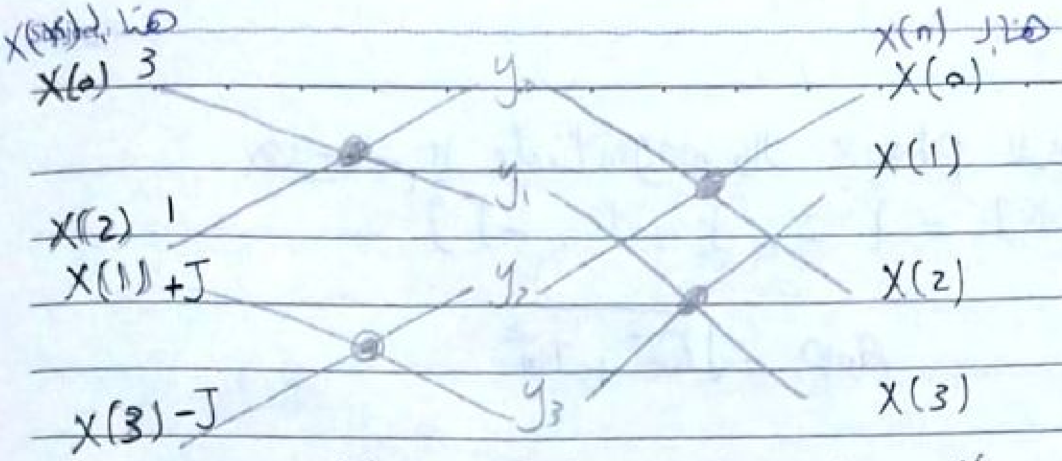
$$\rightarrow X(2) = \Big[1 + 0 + \overset{1}{\cancel{e^{-J2\pi}}} + \overset{-1}{\cancel{e^{-J3\pi}}}\Big] = 1$$

at $k=3$

$$\rightarrow X(3) = \Big[\cancel{1} + 0 + \overset{\cancel{1}}{\cancel{e^{-J3\pi}}} + \overset{-J}{\cancel{e^{-J\frac{9\pi}{2}}}}\Big] = -J$$

نفس نتائج طريقة ال FFT يبقى كل صح بعضه

تعالو بقى نرجعها بال IFFT



$y_0 = X(0) + X(2) = 3 + 1 = 4$

$y_1 = X(0) - X(2) = 3 - 1 = 2$

$y_2 = X(1) + X(3) = J - J = 0$

$y_3 = X(1) - X(3) = J - (-J) = 2J$

~~spectr~~ (x(n)) نجيب بقى

$N\,x(0) = y_0 + y_2 \rightarrow x(0) = \frac{y_0 + y_2}{4} = \frac{4 + 0}{4} = 1$

ناخد بالك من

$N\,x(1) = y_1 + Jy_3 \rightarrow x(1) = \frac{y_1 + Jy_3}{4} = \frac{2 + \overset{-2}{J * 2J}}{4} = 0$

$N\,x(2) = y_0 - y_2 \rightarrow x(2) = \frac{y_0 - y_2}{4} = \frac{4 - 0}{4} = 1$

$N\,x(3) = y_1 - Jy_3 \rightarrow x(3) = \frac{y_1 - Jy_3}{4} = \frac{2 - \overset{-2}{J * 2J}}{4} = 1$

$\therefore x(n) = \{1, 0, 1, 1\}$

كده الحل بتاعى صح وحلينا كده أول وتانى مطلوب

→ check parseval بعده هنتأكد من حل

$\sum x^2(n) = \frac{1}{N} \sum X^2(K)$

$1 + 0 + 1 + 1 = 3$ .... $\frac{1}{4}[9 + 1 + 1 + 1] = \frac{12}{4} = 3$

تربيع معامل J

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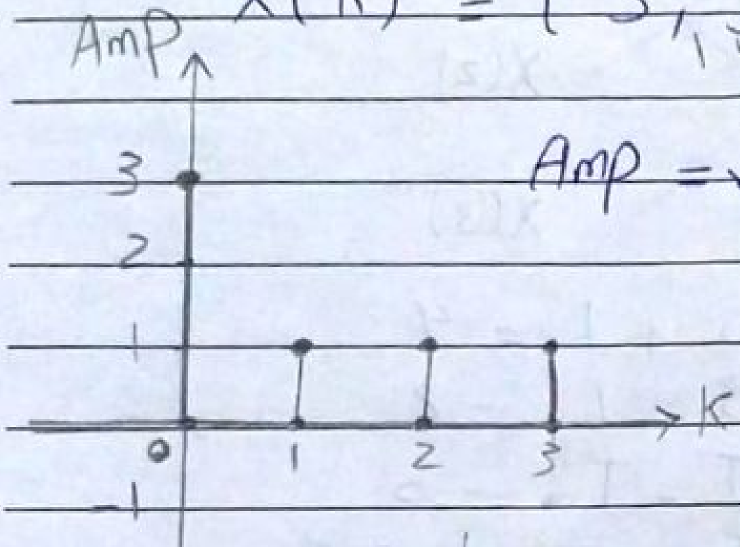
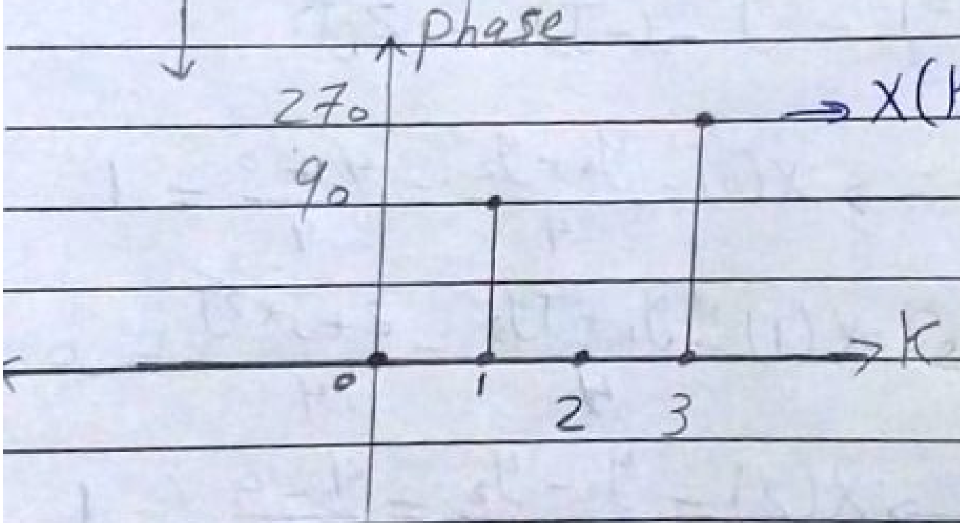
هنرسم ال magnitude وال phase لل spectrum

$X(k) = \{3, \underset{1}{J}, 1, \underset{1}{-J}\}$

$Amp = \sqrt{Re^2 + Im^2}$ ← magnitude



ال phase → $X(k) = \{0, 90, 0, 270\}$



آخر مطلوب هكتب المقارنة بين ال FFT، ال DFT

	No of addition	No of multiplication
DFT	$N(N-1)$	N^2
FFT	$N\log_2 N$	$\frac{N}{2}\log_2 N$